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QoE Prediction from Encrypted Traffic with MARINA on P4/Tofino

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Elmer Dema
March 2026

"You know, Quasimodo predicted all of this"
— B.B.

Abstract

Quality of Experience (Quality of Experience (QoE)) is a crucial metric for multimedia services, as it directly reflects user satisfaction with platforms such as YouTube or Netflix. However, measuring QoE is challenging due to its subjective nature and the difficulty of obtaining real-time user feedback. Traditional methods relying on surveys or manual inputs are not scalable for large-scale network environments.

While papers, such as ViCrypt , have demonstrated that network statistics can predict user QoE, its reliance on end-host GPUs for processing features makes it incompatible for direct implementation in the network data plane [1]. To bridge this critical gap, this thesis investigates the possibility of implementing Machine Learning models for QoE prediction directly within the network data plane to provide real-time monitoring of encrypted video traffic. This work hypothesizes that by using time-windowed rolling statistics and mapping a constrained Random Forest classifier into Ternary Match-Action Tables, it is possible to perform accurate, line-rate QoE prediction strictly within the physical memory and pipeline constraints of the Intel Tofino Application-Specific Integrated Circuit (ASIC).

To achieve this, a hybrid architecture is introduced for real-time QoE monitoring that performs line-rate feature extraction and prediction. This design divides the predictive model into two components: a module deployed on a P4/Tofino switch performs initial video flow classification and extracts primitive features at line rate, while the final QoE prediction is made using a lightweight machine learning model on the switch itself. The main contributions include a mapping of Random Forest classifiers to P4 match-action tables and a comprehensive benchmarking framework for evaluating in-network Machine Learning (ML) performance.

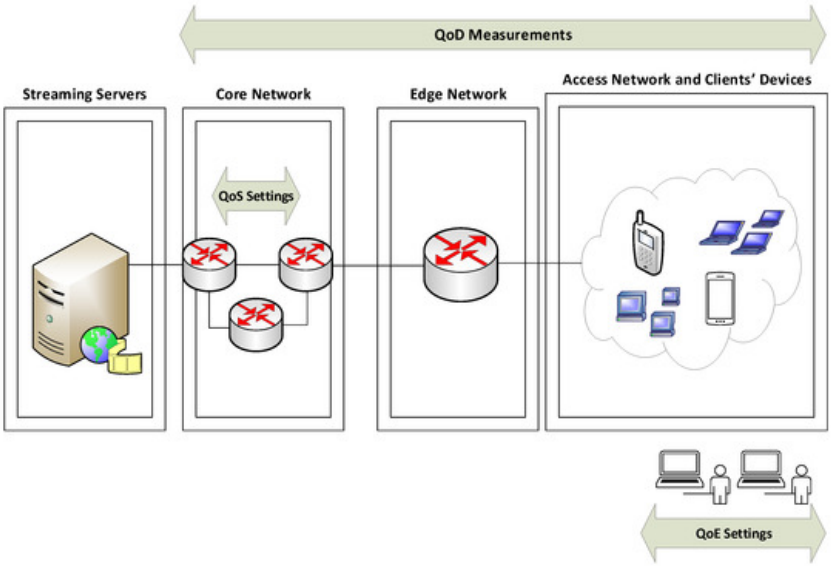

Figure 1: **a network architecture diagram illustrating the data flow from streaming servers to client devices, while highlighting specific stages for measuring Quality of Service (QoS), Quality of Delivery (QoD), and Quality of Experience (QoE). Source**

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1. Introduction

Machine Learning (ML) is a subgroup of Artificial Intelligence (Artificial Intelligence (AI)), a field where computers autonomously learn from past data to generate predictions or solutions. Recently, there has been a surge in applying ML methods to address practical challenges within computer networks [2]. This thesis implements and extends recent applications of ML techniques that use Quality of Experience (QoE) metrics to predict video quality. These QoE measurements are essential for capturing the end-to-end performance of network services [3]. As video data makes up the majority of global Internet Protocol (IP) traffic [4], network operators are under pressure to optimize resource management while satisfying user demands. In general, Video Quality Assessment (Video Quality Assessment (VQA)) has relied on both subjective [5] and objective [6] methodologies. Subjective methods focus on quantifying the end-user's satisfaction. Consequently, network traffic management driven by Quality of Experience (QoE) depends on monitoring and forecasting application-level performance or QoE via video Key Performance Indicators (Key Performance Indicators (KPIs)) that impact the user. The International Telecommunication Union (International Telecommunication Union (ITU)) characterizes QoE as the overall experience of a service from the subjective viewpoint of the end user [7].

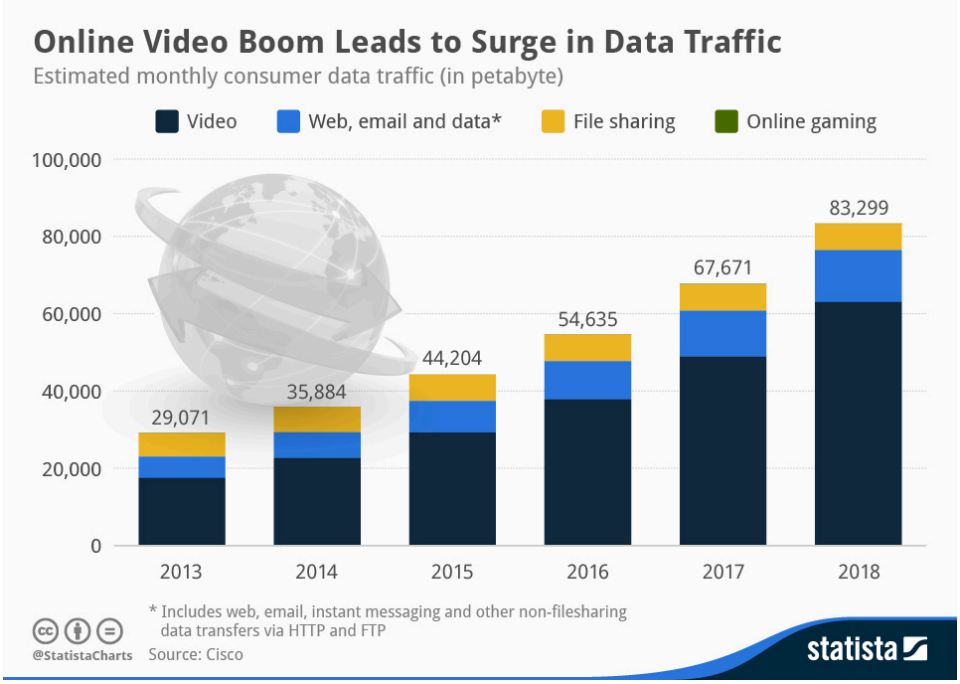

Figure 2: **Video effect of the devices on traffic is more pronounced because of the introduction of Ultra-High-Definition (Ultra-High-Definition (UHD)), or 4K, video streaming. Source**

A significant obstacle in video streaming is the absence of a standardized method[1] for measuring QoE [8], which has stimulated research into QoE models derived from network statistics.

Such modeling assists in defining specific KPIs for various user categories. For example, network providers might analyze tools like rebuffering or quality switching to understand their impact on the video stream.

This helps analyze the effect on user QoE and correlates these issues with network parameters such as jitter or delay. While subjective studies offer reliable evaluations, they are noted for being expensive, time intensive and impractical for real-world applications [9]. Meanwhile, objective VQA uses mathematical models to estimate perceived quality. Although metrics such as Peak Signal-to-Noise Ratio (Peak Signal-to-Noise Ratio (PSNR)) [10] and Structural Similarity Index (Structural Similarity Index (SSIM)) are computationally efficient, they do not always accurately mirror the actual user experience [11], [12].

The main focus in this thesis is on network performance rather than direct user experience. Because the proposed system operates directly within the switch, the focus is placed on network-layer parameters rather than application-layer metrics, with QoE estimated from buffering-related events (e.g., stalling and depletion).

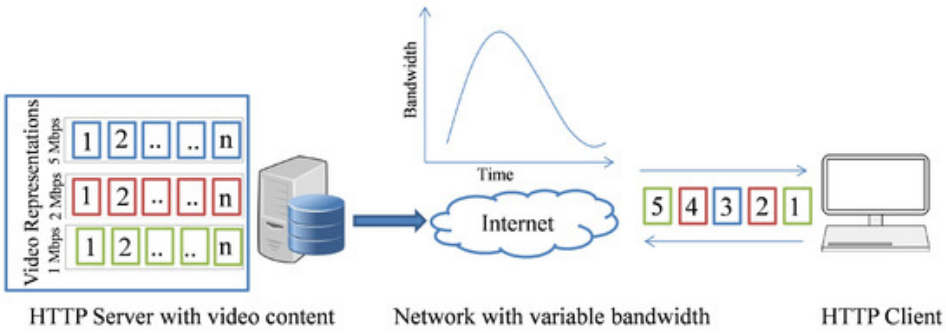

Figure 3: **MPEG DASH is a scenario where a video player dynamically switches between different quality levels of media segments delivered over HTTP to adapt to fluctuating network conditions for smooth playback.**

[1]The P.1203 model was validated only for certain content conditions (e.g., video up to 1080p and 25 fps). Extrapolating beyond these ranges (e.g., 4K, high fps content) may create unreliable quality estimates unless updated or extended versions are used.

2. Related Work

2.1. Quality of Experience Modeling

Traditional approaches to QoE modeling have relied on subjective user studies and objective mathematical models to quantify end-user satisfaction [6], [7]. Subjective methods, while considered the gold standard for accuracy, are expensive, time-intensive, and fundamentally impractical for deployment in real-world, large-scale network environments [9] . Objective metrics such as Peak Signal-to-Noise Ratio (PSNR) [10] and the Structural Similarity Index (SSIM) [11] offer computational efficiency but have been shown to correlate poorly with actual human perception, particularly under complex network impairment conditions [12].

On the other hand, with the adoption of end-to-end encryption (e.g., HTTPS, QUIC) for multimedia delivery, Deep Packet Inspection (DPI) and payload-based video quality analysis are no longer viable. Consequently, recent literature has pivoted towards inferring QoE metrics directly from encrypted network traffic patterns. A significant challenge in this space is the absence of a universally accepted QoE measurement standard [8]; for instance, the ITU-T P.1203 model was validated only for specific content conditions, limiting its applicability to modern high-resolution streams.

Frameworks such as ViCrypt [1] have demonstrated the feasibility of using Machine Learning (ML) to predict video QoE from network-layer features, establishing a strong baseline for passive, encrypted traffic monitoring. Further extending this, contemporary studies have utilized ML models to estimate key QoE indicators for encrypted DASH traffic in modern cellular networks [13] and live video conferencing platforms [14]. Research by Izima et al. [2] has further explored QoE prediction under time-varying network loads, highlighting the importance of temporal dynamics in feature design. These works collectively establish that statistical network properties such as packet sizes, inter-arrival times (Inter-Arrival Time (IAT)), and jitter are strong predictors of application-level events like buffer depletion and stalling, forming the theoretical basis for the feature engineering in this thesis.

2.2. In-Network Machine Learning

While ML-based QoE prediction is well-established, relying on end-host GPUs or external collector servers introduces non-trivial processing latency and limits scalability in high-throughput environments. Recent advancements in programmable data planes, specifically using the P4 language [15], have enabled the displacement of some monitoring and decision-making logic from the control plane to the data plane [16]. This paradigm shift allows classification and telemetry tasks to execute at line rate, directly within the forwarding hardware, eliminating the round-trip cost of offloading decisions to a separate server.

Executing complex ML models like Random Forests at line rate, however, requires specialized hardware mapping, as programmable ASICs such as the Intel Tofino do not natively support recursive branching or floating-point arithmetic. The pForest architecture [17] introduced the foundational concept of translating context-dependent decision trees into Match-Action Tables (Match-Action Tables (MATs)) for in-network inference,

demonstrating that tree traversal can be unrolled across sequential pipeline stages. Building upon this paradigm, frameworks like Planter [18] have formalized and automated the mapping of a broad range of ensemble ML models — including Random Forests, gradient boosted trees, and support vector machines — into programmable switches, significantly lowering the barrier for in-network ML deployment. Other recent works have explored resource-efficient implementations of tree-based models using feature sharing and knowledge distillation [19], reducing TCAM pressure by sharing common decision nodes across trees. Hybrid approaches, such as IIsy [20], partition the classification task between the data plane and a co-located control-plane agent, relaxing hardware constraints while preserving low-latency inference for the majority of traffic. These methodologies collectively provide the architectural foundation for our approach of translating a unified Random Forest model into unrolled physical pipeline stages using Ternary Content-Addressable Memory (TCAM).

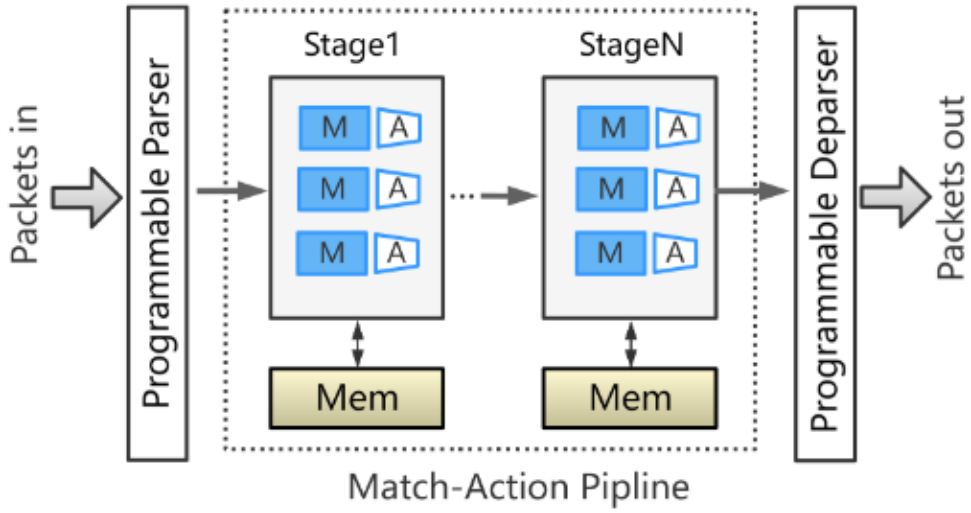

Figure 4: **The general architecture of a P4 switch. [19]**

2.3. Benchmarking and Hardware Constraints

Evaluating the performance of in-network ML models requires a rigorous understanding of the strict physical constraints imposed by programmable ASICs. Comprehensive surveys on in-network machine learning [21] have characterized the fundamental bottlenecks of devices such as the Intel Tofino: available SRAM and TCAM capacity, Very Long Instruction Word (VLIW) instruction slot budgets, and a hard limit on the number of physical pipeline stages. These constraints are not merely engineering inconveniences; they define a hard feasibility boundary that determines whether a given model can be compiled and deployed at all.

Studies evaluating hybrid in-network classification frameworks, such as IIsy [20], emphasize the complex, multi-dimensional trade-offs between a model's predictive accuracy and its hardware resource footprint, showing that naive model translation routinely exceeds available pipeline resources. Furthermore, because the Tofino architecture enforces a strict Directed Acyclic Graph (Directed Acyclic Graph (DAG)) for table execution, the recursive branching inherent in standard decision trees must be

aggressively flattened before compilation [22]. Feature selection therefore plays a dual role: beyond its conventional impact on statistical performance, it directly determines the TCAM and VLIW instruction budget consumed by each pipeline stage. Recent work by Xie et al. [19] has demonstrated that knowledge distillation can compress decision tree ensembles into shallower structures that respect these stage limits without significant accuracy loss. These findings directly motivate the benchmarking methodology adopted in this thesis, in which feature count, tree depth, and ensemble size are systematically swept to map the feasibility boundary on Tofino 2 hardware.

3. Design & Methodology

3.1. REPORTER System Architecture

The foundation of the solution builds upon the Reporter[2] framework, a programmable switch architecture designed for real-time traffic classification. As illustrated in Figure 5, the system operates on a decoupled **Control Plane** and **Data Plane** model to handle high-throughput video traffic.

The processing flow begins in the Data Plane when raw packets enter the ingress pipeline. For every incoming packet, the REPORTER updates Feature Registries with new telemetry data. A subset of these extracted features is then used by a **Random Forest (Random Forest (RF)) Classifier**, which is encoded directly into the reporter's hardware. This classifier determines if a packet belongs to a video flow.

If a packet is classified as a video packet, the REPORTER sends a **telemetry packet** to a downstream Translator Switch at a specific time interval. This telemetry packet carries the extracted features and the classification result. The Translator Switch then receives this packet and uses it to perform further **Quality of Experience (QoE) classification** by running a specific, potentially different, model on the incoming classified video packets.

In parallel, the Control Plane oversees model lifecycle management. It handles the offline training of Random Forest models, encodes them into hardware table entries via the `TofinoForestManager`.

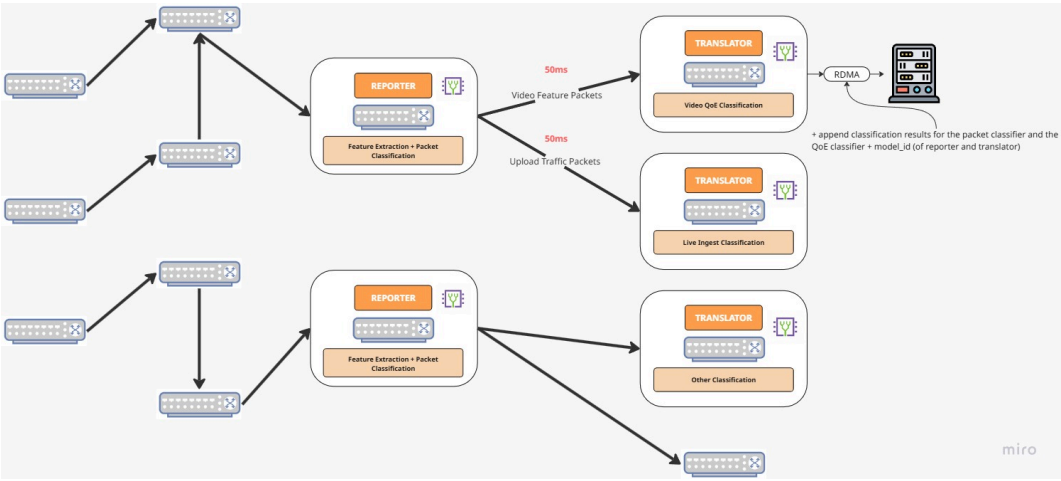

Figure 5: **Architecture of the REPORTER framework, showcasing the interaction between Control Plane model management and Data Plane telemetry generation.**

[2]Data-Plane Telemetry Architecture (DTA) is a system consisting of several components, each in their own directories.Reporter is a DTA reporter switch. This switch can generate telemetry reports through DTA."

3.2. Data Plane Translator Architecture

The "Translator" is the intelligent core of the proposed in-network telemetry system. While the Reporter component is responsible for aggregating flow statistics, the Translator consumes these statistics, executes a machine learning inference directly within the data plane, and enforces state-dependent routing policies.

Because standard decision trees rely on recursive branching (`if-else` structures) which are incompatible with the strict pipeline constraints of the Intel Tofino ASIC, the P4 program was automatically generated using a custom script (`p4_generator_tna.py`). This generator translates a trained Random Forest model into hardware-compatible Match-Action Tables (MATs).

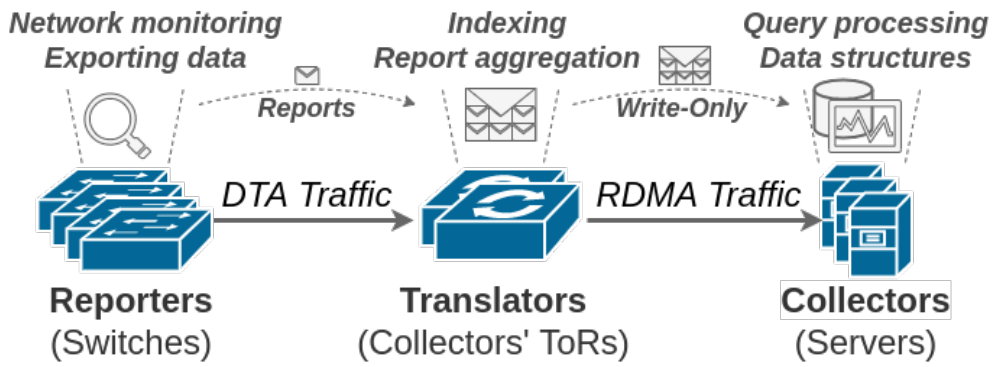

Figure 6: **Translator is a DTA translator switch. This switch will intercept DTA reports and convert these into Remote Direct Memory Access (RDMA) traffic. It is in charge of establishing and managing RDMA queue-pairs with the collector server. Source**

3.2.1. Telemetry Parsing and Feature Extraction

The Translator operates primarily on Data-Plane Telemetry Architecture (DTA) packets. Upon ingress, the custom `SwitchIngressParser` inspects the packet headers. If it detects a User Datagram Protocol (UDP) packet destined for port `40040` with the Marina opcode (`0x05`), it transitions into specialized parsing states to extract the `dta_keyval_static_h` and `keywrite_data_h` headers.

Once parsed, the `FeatureExtraction` control block maps the payload into the `local_metadata_t` structure. While features like `packet_count`, `ps_sum`, and `iat_sum` are extracted directly from the payload, the Translator dynamically calculates `jitter` on the fly. It does this by subtracting the flow's `last_packet_timestamp` (carried in the DTA header) from the switch's local hardware ingress timestamp (`ig_intr_md.ingress_mac_tstamp`).

3.2.2. Random Forest Hardware Mapping

To execute a Random Forest within the Tofino data plane, the `p4_generator_tna.py` script employs two critical design patterns: Ternary Match-Action mapping and Physical Stage Unrolling.

3.2.2.1. Ternary Range Matching (TCAM)

Decision tree nodes evaluate features using inequalities (e.g., $X_{\text{jitter}} < 500$). The Tofino ASIC evaluates inequalities using Ternary Content-Addressable Memory (TCAM). The generator maps the decision thresholds into ternary bitmasks.

Each tree is represented by a set of tables where the match keys include the `current_node` (Exact match) and the four network features (`packet_count`, `ps_sum`, `iat_sum`, `jitter`) as Ternary matches. Depending on the matched rule, the ALU executes either `action_goto_node` to traverse deeper into the tree, or `action_set_leaf` to output a class prediction.

```p4
table tbl_tree_0_stage_0 {
    key = {
        local_md.classification_variables.current_node_tree0: exact;
        local_md.features.packet_count: ternary;
        local_md.features.ps_sum: ternary;
        local_md.features.iat_sum: ternary;
        local_md.features.jitter: ternary;
    }
    actions = {
        action_goto_node_tree_0;
        action_set_leaf_tree_0;
        NoAction;
    }
}
```

Listing 1: P4 representation of a decision tree node using ternary matching for feature threshold evaluation.

3.2.2.2. Physical Stage Unrolling

The Tofino architecture enforces a strict Directed Acyclic Graph (DAG) for table execution; a table cannot recursively call itself, nor can multiple tables modifying the same metadata execute in the same physical pipeline stage.

To bypass this, the generator "unrolls" the trees. If the Random Forest relies on trees with a maximum depth of 3, the generator creates three distinct tables per tree (e.g., `tbl_tree_0_stage_0`, `tbl_tree_0_stage_1`, `tbl_tree_0_stage_2`). In the `apply` block, these tables are executed sequentially, allowing the packet to traverse from the root to the leaf node across consecutive physical ALU stages.

3.2.3. Voting Mechanism

Because a Random Forest consists of multiple independent decision trees (in this implementation, four trees), their individual predictions must be aggregated to form a final classification. This is handled by a dedicated `tbl_voting` match-action table.

To conserve limited Arithmetic Logic Unit (ALU) cycles, the voting logic is pre-computed by the Python generator and hardcoded as exact-match entries. The table takes the four leaf outputs (`tree0_result` through `tree3_result`) as exact match keys and outputs the majority `final_class` alongside a confidence score (the vote tally).

3.2.4. State-Dependent Routing and Deparsing

Once the `final_class` is resolved, the Translator uses this result to perform Quality of Experience (QoE) or security routing. This is controlled by the `tbl_classification_action` table.

If the traffic is classified as benign (Class 0), the switch executes `forward_benign` and routes the traffic along its normal path (e.g., egress port 36). However, if the traffic is classified as critical or malicious (Class 1), the switch can execute dynamic mitigation strategies, such as `forward_to_ids` (redirecting the packet to port 64 for deep packet inspection) or `drop_malicious` to quarantine the flow entirely at line rate.

Finally, during the Deparsing stage, the switch does not strip the telemetry. Instead, it recomputes the IPv4 checksum and emits the packet along with a newly appended `classification_h` header. This custom header encapsulates the final class result, the confidence score, and the exact tree vote distribution, allowing downstream analytics servers to verify the Tofino ASIC's in-network inference accuracy.

3.3. Evolution of the Methodology

The initial phase of this research focused on establishing a baseline classification model using MARINA statistical features [16], where packet sizes and inter-arrival times were aggregated into isolated 50ms time windows using sums and moments. This earlier approach treated every time window as an independent event, relying on a simplistic labeling strategy that categorized network states solely based on the slope of buffer changes (filling versus depleting) without regard for the absolute buffer level.

While this established a foundational correlation between traffic volume and buffer trends, the model lacked temporal context. It could not distinguish between a sudden, transient traffic drop and a sustained outage, nor could it differentiate between a benign buffer drop from a high level and a critical depletion event leading to a video stall.

To address these limitations, the updated pipeline restructured the feature engineering process to introduce **temporal awareness** and stateful logic suitable for deployment on network switches. This shift transitioned the system from a stateless traffic analyzer to a predictive, risk-aware QoE monitor.

3.3.1. Advanced Feature Engineering: Rolling Statistics

A raw 50ms observation is inherently noisy; a single dropped packet or a micro-burst can cause a spike in jitter that does not necessarily reflect the true network state. To smooth these fluctuations and capture the underlying trends (velocity and acceleration) of the traffic, the feature set was augmented using Exponential Moving Averages (Exponential Moving Average (EMA)).

EMA is particularly suitable for resource-constrained hardware, such as P4-enabled switches, because it allows for "rolling" trend analysis without requiring the storage of a large historical buffer. The switch only needs to store the previous EMA value. The EMA at time t is calculated as:

$$\mathrm{EMA}_t = \alpha \cdot X_t + (1 - \alpha) \cdot \mathrm{EMA}_{t-1}$$

Formula 1: **Exponential Moving Average formula, where X_t is the current observation and α is the smoothing factor.**

Where X_t is the current observation (e.g., jitter or throughput) and α is the smoothing factor. By calculating the EMA over a 20-window span (approx. 1 second), the model gains historical context, allowing it to react to sustained trends rather than transient noise.

Derived Feature	Mathematical Definition
EMA (Trend)	$\mathrm{EMA}_t = \alpha \cdot x_t + (1-\alpha) \cdot \mathrm{EMA}_{\{t-1\}}$
Lag (Immediate History)	$x_{\mathrm{prev}} = x_{\{t-1\}}$
Rate of Change	$\Delta_{\mathrm{bw}} = \mathrm{BWE}_t - \mathrm{BWE}_{\{t-1\}}$

Table 1: **Temporal features engineered to capture historical context and trends.**

3.3.2. Target Class Definition: Risk-Based QoE State

A supervised learning approach requires defined target labels. The existing approach defines a binary classification scheme. A "Lookahead" mechanism was implemented to inspect the buffer state 10 steps (500ms) into the future. By comparing the future buffer level ($B_{\{t+10\}}$) with the current level (B_t), a slope was calculated.

The 'At_Risk' class is defined as a depleting trend occurring when the buffer is dropping by more than a set threshold. This allows the model to prioritize the detection of QoE violations over simple fluctuations.

```python
LOOKAHEAD_STEPS = 10       # Look 500ms into the future
DROP_THRESHOLD = -200      # Buffer dropping by > 200ms

# Binary label: At_Risk if buffer is dropping significantly, Steady otherwise
df['qoe_state'] = np.where(
    (df['buffer_slope'] < DROP_THRESHOLD),
    'At_Risk',
    'Steady'
)
```

Listing 2: Logic used to define the ground-truth QoE target classes.

The resulting class distribution represents `At_Risk` instances dominating the dataset (61.3%) compared to `Steady` stable samples (38.7%).

3.4. Preliminary Modeling and Feature Importance

To establish a baseline for classification performance, a Random Forest Classifier was trained using 150 estimators and a maximum depth of 12. To mitigate the extreme class imbalance, the model utilized `class_weight='balanced'`, which penalizes misclassification of the minority class ("Steady") more heavily.

The input feature vector X for this baseline experiment was reduced to four core metrics: `ps_sum`, `ps2_sum`, `ps3_sum`, and `jitter`.[3]

[3] This initial experiment focused on core traffic metrics to establish a performance baseline.

3.4.1. Performance Evaluation

The model was evaluated on a stratified test set (25% split). As shown in the classification report, the model demonstrated accuracy of 0.76, correctly predicting the 'At_Risk' class with an F1-score of 0.80 and the 'Steady' class with an F1-score of 0.69.

Class	Precision	Recall	F1-Score
At_Risk	0.80	0.80	0.80
Steady	0.69	0.69	0.69

Table 2: **Random Forest performance metrics for QoE state classification.**

3.4.2. Feature Importance Analysis

The Random Forest provided insights into feature relevance. The Gini importance scores revealed that the higher-order statistical moments of traffic volume (`ps2_sum`, `ps3_sum`) and the aggregate traffic volume (`ps_sum`) were the most significant predictors of the buffer's future state, collectively accounting for over 62% of the model's decision-making power.

Feature	Importance Score
Traffic Volume Squared (`ps2_sum`)	0.244
Traffic Volume Cubed (`ps3_sum`)	0.197
Traffic Volume (`ps_sum`)	0.184
Jitter (`jitter`)	0.117

Table 3: **Feature importance ranking derived from the Random Forest model.**

Finally, the processed feature matrix X and the one-hot encoded target variables Y were exported as NumPy arrays (`.npy`) to facilitate ingestion into deep learning frameworks for subsequent experiments.

4. Implementation

4.1. Dataset Analysis

The dataset used in this thesis originates from Karagkioules et al. paper. The dataset is designed to fill this gap by providing measurements that were simultaneously obtained at the network, transport, and application layers. The data were generated using YouTube's native Android application on two smartphone models at two locations in Europe over a period of more than five months.

At the application layer, a wide range of adaptive streaming parameters was extracted from YouTube's mobile client. This was made possible by a recently introduced Wrapper App [23], which enables remote control and monitoring of YouTube's native Android application. At the transport and network layers, the commonly used tool tcpdump [24] was used to record unfiltered packet logs on both the smartphones and a gateway.

The dataset includes measurements from 8 different scenarios, each representing a different quality and rate limiter. However, Scenario 6 is used since it provides automatic quality selection without any rate limiting, which closely resembles real-world conditions. [25]

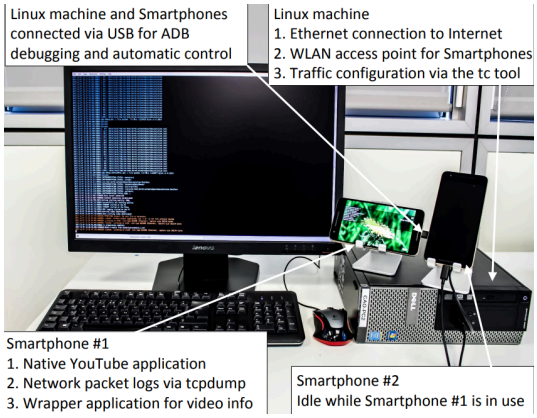

Figure 7: **Hardware setup for regulated and automatic HAS traffic measurements**

4.1.1. Data Ingestion and Parsing

```
(15174 IP(tos 0x0,ttl 55,id 0,offset 0,flags[DF],protoUDP(17),length
1378)74.125.105.91.443>192.168.10.200.4864: UDP, length 135)
```

Listing 3: Sample line from TCPdump network trace log showing a packet with timestamp and length.

The raw data for "Scenario 6" was distributed across a directory structure organized by Video ID and Iteration number. Each iteration contained two distinct data sources: network traffic logs (TCPdump) and application performance metrics (Phone Statistics). To ingest this data, custom parsing functions were developed to handle the semi-structured nature of the logs.

For the network traces, a regular expression was used to extract the timestamp and packet size (length) from each line of the TCPdump output. The regex pattern used to capture these groups is defined below:

```
r'(?P<timestamp>\d+\.\d+).*? length (?P<length>\d+)'
```

Listing 4: Regular expression pattern used to extract packet arrival times and payload lengths.

In contrast, the application logs required a two-step parsing process. The parser first identified lines containing JSON objects embedded within the log text. It then extracted the JSON string, corrected formatting inconsistencies (such as trailing braces), and parsed the object to retrieve the ground-truth labels: Bandwidth Estimate (`bwe`) and Buffer Level (`buffer_level_ms`).

4.1.2. Synchronization and Preprocessing

A critical challenge in this pipeline was aligning the timelines of the two data sources. The application logs recorded timestamps in a timezone differing from the network traces. To rectify this, a one-hour offset was added to the application statistics indices ($t_{\text{stats}} + 1\text{h}$).

Furthermore, to ensure that the feature set captured the context relevant to the video session, the network packets were filtered to a specific time range. This range was defined dynamically as the interval $[t_{\min} - 10\text{s}, t_{\max} + 10\text{s}]$, where $t_{\min}$ and $t_{\max}$ represent the start and end of the application logging period.

4.1.3. Feature Engineering and Windowing

The transformation from raw packet logs to model-ready features involved computing instantaneous metrics followed by a temporal aggregation. The system used a non-overlapping time window of **50ms**.

First, specific metrics were calculated for every packet i in the stream. The Inter-Arrival Time (IAT) was derived as the difference in timestamps between consecutive packets. Jitter was calculated as the absolute difference between consecutive IAT values. To capture the higher-order statistical properties of the traffic flow, the second and third moments (squares and cubes) were computed for both packet size (PS) and IAT.

$$\text{IAT}_i = t_i - t_{i-1}$$

Formula 2: **Inter-Arrival Time between consecutive packets $i - 1$ and i, where t_i is the arrival timestamp of packet i.**

$$J_i = |\text{IAT}_i - \text{IAT}_{i-1}|$$

Formula 3: **Instantaneous jitter, defined as the absolute difference between consecutive Inter-Arrival Times.**

$$J_i = J_{i-1} + \frac{|\mathrm{IAT}_i - \mathrm{IAT}_{i-1}| - J_{i-1}}{16}$$

Formula 4: **RFC 3550 running jitter estimate, which provides a smoothed, exponentially weighted approximation of jitter as used in RTP streams.**

$$|(J) = \frac{1}{N}\sum_{i=1}^{N}|\mathrm{IAT}_i - \mathrm{IAT}_{i-1}|$$

Formula 5: **Mean jitter over a window of N packets, representing the average deviation of packet spacing.**

$$M_k(\mathrm{PS}) = \sum_{i=1}^{N}\mathrm{PS}_i^k, \quad k \in \{2,3\}$$

Formula 6: **Second and third statistical moments of packet size within a time window, capturing traffic burstiness and asymmetry.**

$$\mathrm{Throughput}_t = \frac{\sum_{i\in t}\mathrm{PS}_i}{\Delta}t$$

Formula 7: **Estimated throughput per time window t, computed as total byte volume divided by the window duration Δt.**

These values were then resampled into 50ms buckets. The aggregation rules applied to generate the final feature vector X_t for each window t are detailed in Table 1.

Feature Name	Mathematical Derivation	Aggregation Method
Packet Count	Count of events in window	Sum
Traffic Volume (PS)	$\sum \mathrm{PS}_i$	Sum
Traffic Moments	$\sum \mathrm{PS}^2, \sum \mathrm{PS}^3$	Sum (of squares/cubes)
IAT Total	$\Delta t = t_i - t_{i-1}$	Sum
IAT Moments	$\sum \mathrm{IAT}^2, \sum \mathrm{IAT}^3$	Sum (of squares/cubes)
Jitter	$\lvert\mathrm{IAT}_i - \mathrm{IAT}_{i-1}\rvert$	Mean (Sum / Count)

Table 4: **Definition of features extracted per 50ms time window.**

Statistic	Jitter(ms)	Buffer Level (ms)
Count	163,090	163,090
Mean	40.78	105,381.84
Std Dev	441.41	29,514.49
50% (Median)	1.2	120,077.00
75% (Q3)	1.78	122,075.75
Max	9,231.11	129,857.00

Table 5: **Summary statistics for Jitter and Buffer Level across the processed dataset.**

4.1.4. Target Alignment and Dataset Construction

The final stage of the pipeline merged the high-frequency network features with the lower-frequency application labels. Since the application statistics (Buffer Level and Bandwidth Estimate (BWE)) were logged at irregular intervals, a standard join was insufficient. Instead, a `merge_asof` strategy was implemented.

For each 50ms feature window, the system located the nearest application log entry. A strict tolerance limit of **100ms** was enforced; if no application label existed within 100ms of the window's timestamp, the sample was discarded. This ensured that the model would not learn from stale state information.

Post-merge, the dataset underwent a cleaning phase where any rows containing `NaN` values for the target variables were dropped. The final output consisted of a CSV file containing approximately 163,000 samples, encompassing the feature vector, the target labels, and identifiers for the video and iteration.

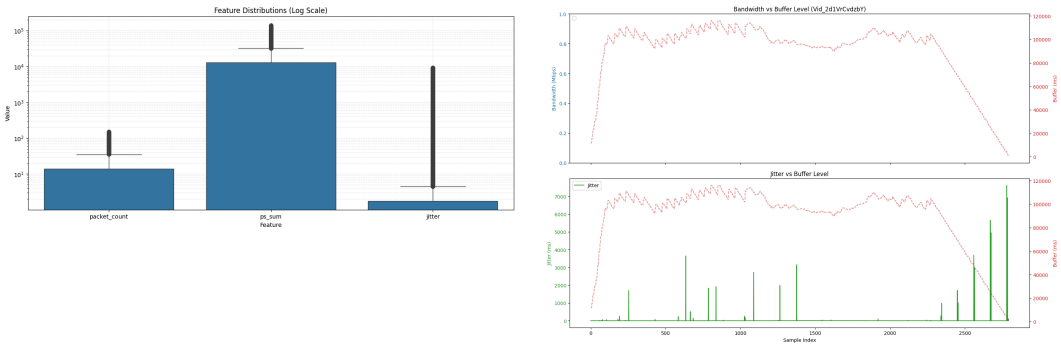

Figure 8: **(Left) Feature distributions (log scale); (Right) Bandwidth vs Buffer Level over time.**

4.2. Model Training via pForest

Following the feature engineering chapter, a simplified version of the pForest [17] architecture was adopted to enable in-network inference on the Intel Tofino switch.

This phase uses the pre-processed datasets (`np_data.npy` (features) and `np_dummies.npy` (labels)) from the previous step. These files contain the statistical snapshots required to train the decision trees.

```terminal
├── p4_generator_tna.py
├── randomforest
│   ├── randomForestEncode.py
│   ├── rf_model.py
│   ├── storage
│   │   ├── np_data.npy
│   │   └── np_dummies.npy
│   ├── tester.py
│   └── voting_table_gen.py
```

Listing 5: Files for the Random Forest training logic

4.2.1. The Training Pipeline (`rf_model.py`)

The training logic is found in the `rf_model.py` script. While it imports the standard `scikit` library, the pipeline is constrained by the hardware limitations of the Tofino architecture, specifically the available Match-Action Table (MAT) resources and SRAM capacity.[4]

To ensure the trained model can be compiled to the P4 data plane, the script performs two critical adaptations:

1. **Hardware-Constrained Feature Selection:** Although the initial dataset contains a rich set of features, the Tofino switch cannot efficiently track high-dimensional state vectors for every flow without exhausting memory. Consequently, the training script explicitly reduces the feature space to the four most impactful metrics identified during the preliminary analysis:
   - Index 1: `ps_sum` (Traffic Volume)
   - Index 2: `ps2_sum` (Traffic Volume Squared)
   - Index 3: `ps3_sum` (Traffic Volume Cubed)
   - Index 7: `jitter`
2. **Hyperparameter Tying:** The model complexity is controlled via `num_trees` and `max_depth`. In this context, `max_depth` dictates the number of pipeline stages required for a decision, while `num_trees` impacts the parallel lookup width.

[4] Intel Tofino 1 has around 10 Mbit of SRAM per stage, while Tofino 2 explicitly includes 16 Mbit per stage.

```python
def build_and_train_random_forest(num_trees, max_depth):
    features = np.load(".../storage/np_data.npy", allow_pickle=True)
    labels = np.load(".../storage/np_dummies.npy", allow_pickle=True)

    # REDUCED: Select only top 4 features to fit Tofino constraints
    # Indices: 1=ps_sum, 2=ps2_sum, 3=ps3_sum, 7=jitter
    feature_indices = [1, 2, 3, 7]
    features = features[:, feature_indices]

    rf = RandomForestClassifier(n_estimators=num_trees,
    max_depth=max_depth)
    rf.fit(train_features, train_labels)
    return rf
```

Listing 6: Excerpt from `rf_model.py` demonstrating the loading of exported NumPy data and hardware-specific feature pruning.

4.2.2. Deviation from Original pForest Architecture

A significant architectural distinction exists between the original pForest code [26] and our implementation. The original paper proposes **context-dependent** Random Forests, where distinct models are trained for different phases of a flow (e.g., one model for the first 5 packets, another for packets 6-10). This allows for highly specialized early detection. [17]

In contrast, our implementation deploys a Single Unified Random Forest. This model is trained on the features of **completed** flows (whole-flow statistics) but is applied to live traffic where features accumulate incrementally.

4.2.3. Automated P4 Code Generation (`p4_generator_tna.py`)

To bridge the gap between the trained model and the hardware implementation, the system uses `p4_generator_tna.py`. This script builds the P4 program, ensuring compliance with Tofino's strict resource constraints,such as Ternary Content Addressable Memory (TCAM) usage and stage limitations.

The pForest script is managed by a single main function, `generate_full_p4_code`. This function accepts the model's structural hyperparameters (`num_trees`, `max_depth`) and sequentially assembles the pipeline components: identifying necessary headers, generating the "physically unrolled" match-action tables for the decision trees, and constructing the static voting logic.

The final output is written to `rf_classifier_dta_marina.p4`, producing a fully compilable network program that integrates the specific Random Forest logic into the switch's standard forwarding behavior.

4.2.4. Control Plane Configuration: Forwarding

In a Software-Defined Networking (Software-Defined Networking (SDN)) architecture using P4, compiling and loading the program onto the Tofino ASIC only defines the pipeline's structure. Upon initialization, the match-action tables within the data plane are entirely empty, meaning the switch will default to dropping all incoming packets. To

establish connectivity within the hardware-in-the-loop testbed, a control plane application is required to populate the forwarding tables.

For this evaluation, a Python script was developed utilizing the Barefoot Runtime (BFRT) gRPC interface. Unlike full deployment scripts that initialize the entire ML pipeline, telemetry thresholds, and routing logic simultaneously, this script was intentionally decoupled to focus exclusively on the `tbl_forward` table.

4.2.5. Isolated Forwarding Table Configuration

The decision to isolate the forwarding logic was driven by the need for modularity during testing. By removing the overhead of configuring the entire P4 pipeline, this script allows for rapid, incremental updates to the routing topology without resetting the switch's telemetry or machine learning states. This is particularly advantageous during debugging or when switching between the "Reporter" and "Translator" testing phases, as it guarantees that basic Layer 3 forwarding remains intact regardless of the data plane's experimental features.

4.2.6. BFRT gRPC Implementation

The script connects to the Tofino's BFRT server on `localhost:50052` and binds to the deployed pipeline profile (`p4_marina_reporter`). It then targets the `SwitchIngress.tbl_forward` table, retrieving the dynamically assigned action IDs directly from the compiler's output, which ensures robustness against P4 code modifications.

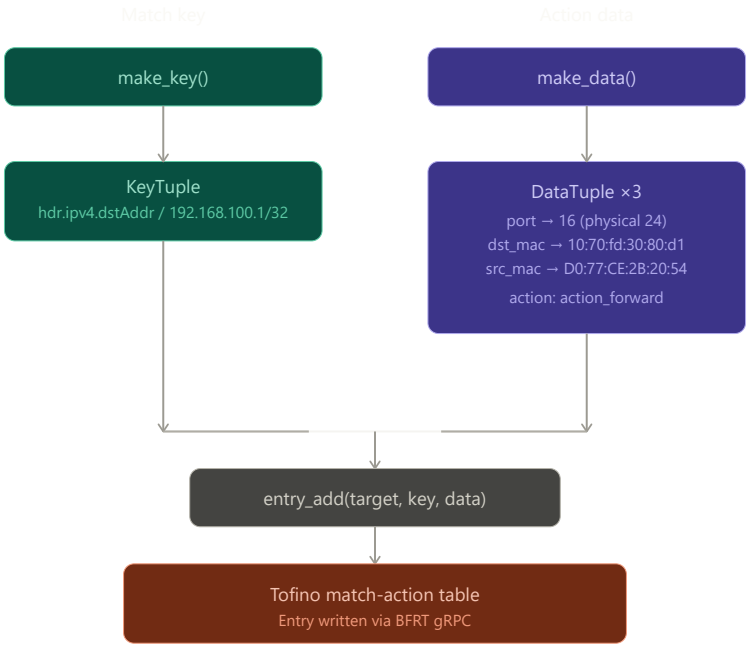

Figure 9: **BFRT gRPC entry insertion flow: a match key (destination IP prefix) and action data (egress port and MAC addresses) are constructed separately and passed to texttt{entry_add()}, which writes the forwarding rule into the Tofino match-action table.**

4.2.7. Table Logic and Topology Integration

The hardware testbed consists of the TRex traffic generator communicating with the Tofino switch via an intermediate Arista switch. Because the TRex server generates traffic destined for `192.168.100.1`, the Tofino switch must correctly identify these packets, assign them to the correct egress port, and rewrite the Ethernet headers to ensure the Arista switch forwards them back to the TRex RX interface.

The inserted entry implements Exact Match logic on the IPv4 destination address. When a packet matches, the action parameters (`port`, `dst_mac`, and `src_mac`) are applied natively at line rate by the switch's ALUs. Table 3 outlines the exact mapping programmed by the control plane script.

Component	Parameter	Configured Value
Match Key	`hdr.ipv4.dstAddr`	`192.168.100.1/32` (Exact Match)
Action Data	`port`	`16` (Logical Egress Port to TRex)
Action Data	`src_mac`	`D0:77:CE:2B:20:54` (Tofino Egress MAC)
Action Data	`dst_mac`	`10:70:fd:30:80:d1` (TRex RX Interface MAC)

Table 6: **Forwarding rules populated in the data plane by the control plane script.**

By successfully executing this script, the control plane ensures that any packet, whether it is raw telemetry traffic or an evaluated packet returning from the QoE ML models, is correctly encapsulated and routed back to the TRex server for statistical analysis and Packet Capture (PCAP) capture. Furthermore, the inclusion of exception handling in the script guarantees that connection failures or missing dependencies are gracefully reported, significantly simplifying the operational workflow of the testbed.

4.2.8. Full Control Plane Script

The following code block contains the complete Python script used to interact with the Barefoot Runtime (BFRT) gRPC interface. This script is responsible for pushing the exact match forwarding rules to the Tofino switch's data plane.

To populate the forwarding plane, a BF Runtime gRPC client connects to the Tofino's control plane server and inserts a single longest-prefix-match entry into `SwitchIngress.tbl_forward`. The entry matches all traffic destined for the configured target IP (prefix length 32) and rewrites both Ethernet addresses before forwarding the packet out the configured egress port toward the TRex receive interface. The table below summarises the fixed parameters used during this experiment.

Parameter	Value	Description
Destination IP	`192.168.100.1/32`	Match key in the forward table
Egress port	`16` (physical 24)	Tofino port toward TRex Rx
Source MAC	`D0:77:CE:2B:20:54`	Tofino port 24 MAC (rewritten)
Destination MAC	`10:70:fd:30:80:d1`	TRex port 1 MAC (rewritten)
gRPC endpoint	`localhost:50052`	BF Runtime control-plane server
Pipeline	`p4_marina_reporter`	P4 program bound at runtime

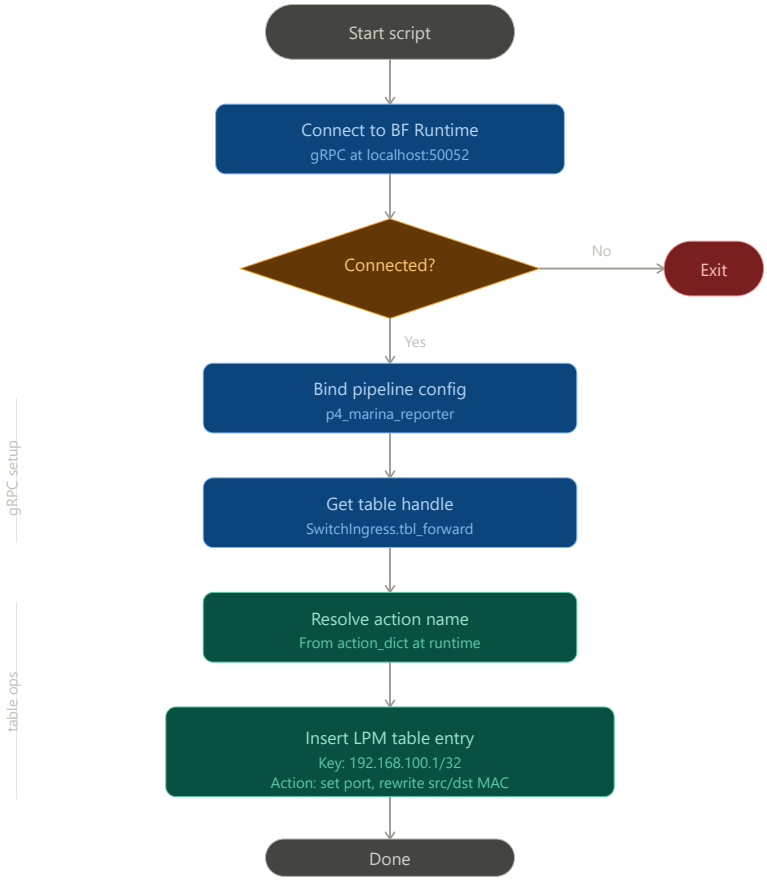

Figure 10: **Control flow of the BF Runtime table insertion script.**

5. Evaluation

5.1. System Evaluation and Translator Testing

To evaluate the performance and accuracy of the in-network Quality of Experience (QoE) classification model, a hardware-in-the-loop testbed was utilized. The topology consists of a TRex software traffic generator connected to an Intel Tofino programmable switch. The evaluation was conducted in a two-phase approach: first generating the necessary telemetry data, and second, testing the Translator's inference and routing capabilities.

5.1.1. Phase 1: Telemetry Data Generation (Reporter)

The Translator component relies on Data-Plane Telemetry Architecture (DTA) packets to perform inference. Because these packets do not exist in the raw public datasets, they had to be synthetically generated using the actual P4 hardware.

To achieve this, the TRex server was configured to replay raw tcpdump logs from scenario 6 of the Würzburg QoE dataset into the Tofino switch. The switch, loaded with the "Reporter" P4 program, processed this traffic, forwarded it back to the server, and generated a dedicated DTA telemetry packet every 100ms containing the aggregated network features. These DTA packets were captured into a PCAP file (`trex_tofino_reply.pcap`) to serve as the ground-truth input for the subsequent Translator evaluation.

Index	Source	Target	Protocol	Size	Info
1	192.168.123.1	192.168.100.1	udp	76	1025 → 4500 Len=34
2	192.168.123.1	192.168.100.1	udp	76	1025 → 4500 Len=34
3	192.168.123.1	192.168.100.1	udp	76	1025 → 4500 Len=34
4	192.168.123.1	192.168.100.1	udp	76	1025 → 4500 Len=34
5	192.168.123.1	192.168.100.1	udp	76	1025 → 4500 Len=34
6	192.168.123.1	192.168.100.1	udp	76	1025 → 4500 Len=34
7	192.168.123.1	192.168.100.1	udp	76	1025 → 4500 Len=34
8	192.168.123.1	192.168.100.1	udp	76	1025 → 4500 Len=34
9	192.168.123.1	192.168.100.1	udp	76	1025 → 4500 Len=34
10	192.168.123.1	192.168.100.1	udp	76	1025 → 4500 Len=34
11	192.168.123.1	192.168.100.1	udp	76	1025 → 4500 Len=34
12	192.168.123.1	192.168.100.1	udp	76	1025 → 4500 Len=34
13	192.168.123.1	192.168.100.1	udp	76	1025 → 4500 Len=34
14	192.168.123.1	192.168.100.1	udp	76	1025 → 4500 Len=34

Figure 11: **Pcap file inspected with pcapviewer extension**

5.1.2. Phase 2: Translator Inference Testing

In the second phase, the Tofino switch was reprogrammed with the "Translator" P4 program, which includes the embedded Random Forest ML tables. The objective was to replay the previously captured DTA packets through the Translator to determine whether the switch could correctly classify the QoE state in real-time, append the classification result to the packet header, and execute state-dependent routing decisions.

To automate this, a custom Python testing framework (`traffic_gen_translator.py`) was developed utilizing the TRex Stateless (STL) API and the Scapy packet manipulation library.

5.1.3. Packet Manipulation and Topology Adaptation

When replaying PCAP files in a live network testbed, the original Layer 2 addresses are no longer valid for the current topology. The script utilizes Scapy to load the DTA packets

(filtered by UDP port 40040) and rewrites the Source and Destination MAC addresses. The Source MAC is updated to the TRex TX interface, and the Destination MAC is set to the Tofino switch.

Crucially, because the IP and UDP headers are modified, the script explicitly deletes the existing checksums, forcing Scapy to recalculate them before transmission. This ensures the packets are not dropped by the switch due to checksum validation failures.

5.1.4. Timing Preservation

Machine learning models analyzing network traffic, especially those utilizing features like jitter, Inter-Arrival Time (IAT), and Exponential Moving Averages (EMA), are highly sensitive to temporal dynamics. Replaying the PCAP file as a single high-speed burst would compress the time windows and invalidate the ML classification.

To prevent this, the script computes the exact microsecond gap between consecutive packets from the original PCAP timestamps. It then utilizes TRex's `STLStream` objects, specifically utilizing the Inter-Stream Gap (`isg`) parameter and the `next` attribute. This creates a chained loop of packets that perfectly mimics the original 100ms emission rate of the Reporter program.

5.1.5. Overcoming Testbed Artifacts: MAC Learning

A significant challenge encountered during testing involved intermediate network equipment (e.g., Arista switches) located between the TRex server and the Tofino switch. Because the TRex RX port only receives traffic and does not natively transmit, the intermediate switch fails to learn the RX port's MAC address, resulting in dropped or flooded return traffic.

To resolve this, a "Keepalive" mechanism was implemented. Before initiating the DTA traffic burst, the script instructs TRex to transmit dummy broadcast UDP packets from the RX port. This forces the intermediate switch to populate its MAC address table (FDB), ensuring a clean and reliable return path for the packets evaluated by the Tofino switch.

```python
# Keepalive on RX port (CRITICAL - prevents intermediate switch drops)
keepalive_pkt = (
    Ether(src=TREX_RX_MAC, dst="ff:ff:ff:ff:ff:ff") /
    #abbr("ip")(src=TREX_RX_IP, dst="255.255.255.255") /
    #abbr("udp")(sport=12345, dport=12345) / Raw(load="keepalive")
)
keepalive_stream = STLStream(
    packet=STLPktBuilder(pkt=keepalive_pkt),
    mode=STLTXCont(pps=5)
)
c.add_streams(keepalive_stream, ports=[rx_port])
c.start(ports=[rx_port], force=True)
```

Listing 7: Implementation of the Keepalive mechanism to trigger MAC learning on intermediate switches.

5.1.6. Execution and Capture

With the topology established and timing configured, the TRex server transitions the RX port into a service mode and begins capturing traffic. The DTA streams are transmitted from the TX port for a defined duration (10 seconds). As the packets pass through the Tofino switch, the P4 Translator program applies the Random Forest classification, appends the resulting QoE state to the packet, and routes it back.

The script continually polls and displays the real-time TX/RX statistics to verify that no packets are dropped during inference. Finally, the test concludes by halting transmission, stopping the capture buffer, and saving the evaluated packets into a new output file (`trex_tofino_translator_reply.pcap`) for offline validation of the model's accuracy.

5.1.7. Telemetry Generation Results

Following the execution of the hardware-in-the-loop tests, the captured traffic (`trex_tofino_reply.pcap`) was parsed offline to verify the proper generation of Data-Plane Telemetry Architecture (DTA) packets by the Tofino switch. The parser processed approximately 285 MB of traffic, successfully filtering and isolating the telemetry reports from the background traffic.

Traffic Metric	Count
Total Captured Packets	20,263
Background / Other Packets	20,228
DTA Telemetry Reports generated	**350**

Table 7: **Summary of the captured PCAP file during the Reporter testing phase.**

Of the 20,2630 packets captured, the switch successfully emitted 350 dedicated DTA telemetry reports (one every 100ms as configured). To validate the hardware logic, the payload of the telemetry reports was fully decoded. Table 2 details the specific fields extracted from Report #1.

The values confirm that the Tofino switch correctly tracked the stateful flow metrics, such as accumulated packet counts, Inter-Arrival Time (IAT) sums, and jitter directly within the data plane. Additionally, the report appends the machine learning classification outcome and a bitmap indicating which specific features were utilized during the Random Forest inference step.

Telemetry Field	Decoded Hardware Value
L2 MAC Address	d0:77:ce:2b:20:54 -> 10:70:fd:30:80:d1
L3 IP Address	192.168.123.1 -> 192.168.100.2
L4 Protocol	UDP 49374 -> 40040 (DTA)
— DTA Header Metadata —	
Opcode	0x05
Redundancy Level	1
Telemetry Key (Flow ID)	2980564395
Monitored Flow	192.168.123.1:1025 -> 192.168.100.1:4500 (UDP)
Packet Count	198
Last Packet Timestamp	850,630,301
Sum of IAT	99,952,039
Sum of IAT^2 / IAT^3	0 / 0
Sum of Packet Size	272,844
Sum of Packet $Size^2$ / $Size^3$	0 / 0
Jitter	2,001,694
Classification Result	**0** (NON-VIDEO)
Used Features Bitmap	0x0000000d

Table 8: **Detailed breakdown of DTA Telemetry Report #1, showcasing the stateful metrics computed by the Tofino hardware and the resulting QoE classification.**

5.2. Benchmarking and Performance Analysis

This section presents the results of the benchmarking framework developed to evaluate the practical limits of in-network QoE prediction. The automated pipeline, which is implemented in the `code/benchmark.py` script, trains Random Forest models with varying hyperparameters, translates them into P4 code, and utilizes the Intel SDE compiler to evaluate architectural resource consumption.

5.2.1. Feature Scaling and Hardware Constraints

The first major experiment evaluates the impact of the number of telemetry features (k) on both Machine Learning (ML) performance and hardware resource utilization. The sweep was conducted using a fixed configuration of 2 trees and depth 2, isolating the effect of feature count from other hyperparameters.

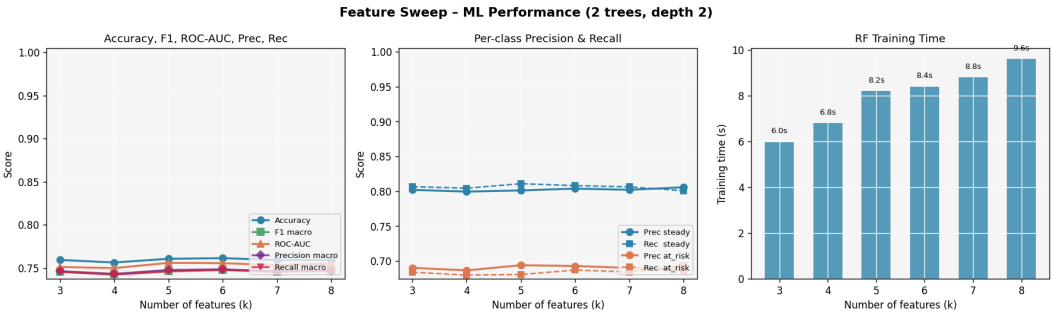

Figure 12: **ML performance metrics across a varying number of features (k).**

As shown in Figure 12, the left panel plots overall accuracy, macro F1, ROC-AUC, macro precision, and macro recall as k is increased from 3 to 8. All metrics remain tightly clustered in the range of 0.75–0.76 across the entire sweep, indicating that the model's predictive power is largely saturated already at $k = 3$. The middle panel breaks this down per class: the *Steady* class consistently achieves a precision and recall of approximately 0.80–0.81, while the *At_Risk* class scores approximately 0.69–0.70 throughout. Critically, neither class shows meaningful improvement as more features are added, confirming that the dominant predictive signal is concentrated in a small subset of traffic statistics. The right panel shows that RF training time increases monotonically from 6.0s at $k = 3$ to 9.6s at $k = 8$, reflecting the additional computational cost of fitting trees over a higher-dimensional feature space. This linear growth in training time, combined with the flat ML performance, strongly motivates restricting the deployed model to the smallest effective feature set.

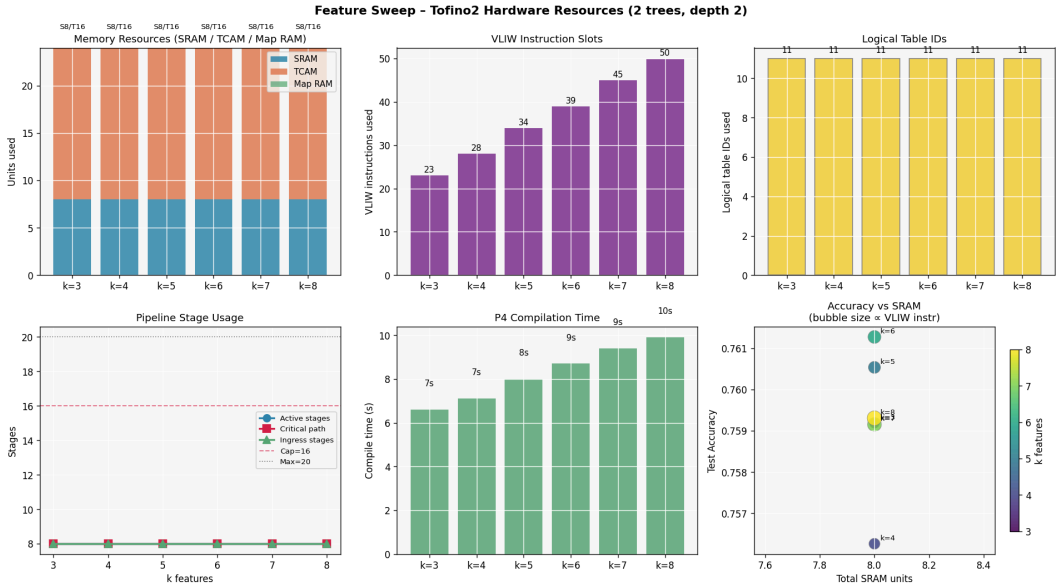

Figure 13: **Hardware resource consumption (SRAM, TCAM, VLIW instructions, and pipeline stages) relative to the number of features.**

Figure 13 reveals the hardware cost of expanding the feature space on the Tofino 2 ASIC. The top-left memory panel shows that SRAM consumption remains constant at approximately 8 units across all values of k, while TCAM usage stays flat at roughly 15 units, both well within the device's S8/T16 budget. The number of logical table IDs is also stable at 11 throughout the sweep, meaning the structural topology of the P4 program does not change with feature count. However, the top-center panel tells a different story for VLIW instruction slots: these grow steeply and linearly from 23 at $k = 3$ to 50 at $k = 8$. Each additional feature introduces new ternary match keys into the decision-tree tables, expanding the instruction word required to evaluate each node. The bottom-left panel confirms that pipeline stage usage remains flat at 8 active stages for all tested values of k, staying well below both the soft cap of 16 and the hard maximum of 20, with compilation times growing moderately from 7s to 10s. The bottom-right scatter plot synthesises the accuracy-versus-SRAM trade-off: all configurations occupy approximately the same SRAM footprint (8.0 units) yet show small but meaningful accuracy differences. Notably, $k = 6$ achieves the highest test accuracy of 0.761 with the largest VLIW footprint (bubble size), while $k = 4$ drops to only 0.757. This analysis confirms that $k = 4$ (the four features ps_sum, ps2_sum, ps3_sum, and jitter selected in the training pipeline) represents the optimal operating point: it delivers near-peak accuracy while minimising VLIW pressure and compilation overhead.

However, this improvement in ML metrics comes at a steep cost in terms of hardware resources. The compiler reports indicate that SRAM, TCAM, and specifically VLIW instruction slots scale rapidly as k increases. The physical limit of 20 pipeline stages on the Tofino 2 architecture becomes a hard bottleneck, as demonstrated in the stage utilization heatmaps.

5.2.2. Tree and Depth Configuration Sweep

The second experiment systematically sweeps the number of trees (3–10) and the maximum depth (d1–d5) of the Random Forest, using the fixed 4-feature set identified in the previous experiment. While larger and deeper forests naturally yield higher test accuracies and better recall, they strain the P4 compiler's ability to fit the logic within the limited hardware stages.

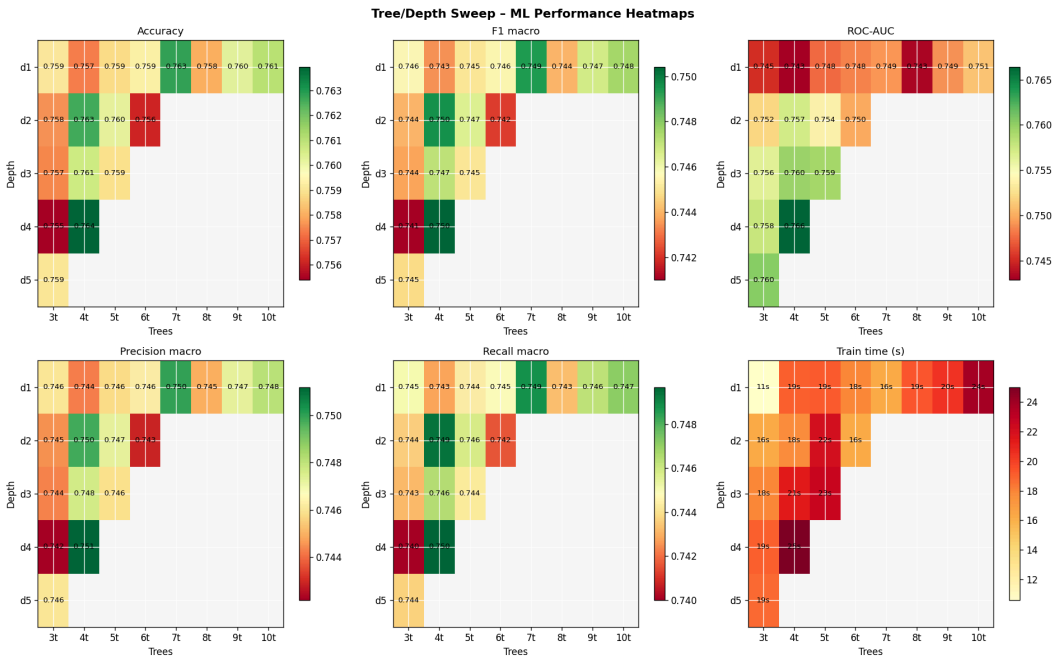

Figure 14: **Heatmaps correlating the number of trees and maximum depth with overall ML performance.**

Figure 14 presents six heatmaps covering accuracy, macro F1, ROC-AUC, macro precision, macro recall, and training time. Cells left blank (white/grey) indicate configurations that failed to compile or were not executed due to hardware infeasibility. Several trends are immediately visible. First, accuracy and F1 scores span a narrow band between approximately 0.756 and 0.763, confirming that the model's predictive ceiling is constrained by the feature set rather than the ensemble size. The highest accuracy of 0.763 is achieved at depth 1 with 7 trees, and at depth 2 with 4 trees. Increasing depth beyond d2 does not improve accuracy and, in fact, produces slight degradations for the configurations that do compile. Second, the ROC-AUC heatmap mirrors this pattern, peaking at 0.765 for depth-1 configurations and declining as depth grows. Third, and most practically significant, the training time heatmap shows that deeper configurations (d3–d5) require substantially more time (up to 21s for d3 with 5 trees), and the compilation grid shrinks dramatically: depth 3 compiles for only 3–5 trees, depth 4 for only 3–4 trees, and depth 5 for only 3 trees. This confirms a fundamental trade-off: marginal accuracy gains from deeper trees are outweighed by the exponential growth in pipeline resource requirements.

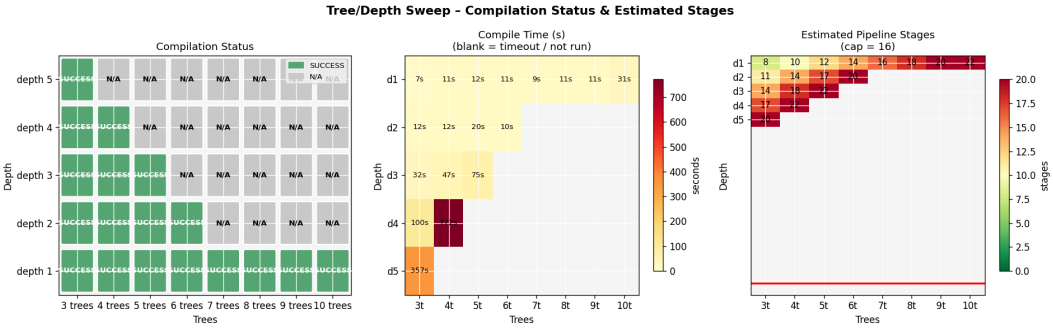

Figure 15: **Compilation status and estimated pipeline stages for various tree and depth configurations. Configurations exceeding the 20-stage limit fail to compile.**

Figure 15 directly maps the hardware feasibility boundary. The left panel shows the compilation status grid: all depth-1 configurations compile successfully across the full range of 3–10 trees; depth-2 succeeds for 3–6 trees; depth-3 for 3–5 trees; depth-4 for only 3–4 trees; and depth-5 only for 3 trees. All other entries are marked N/A, meaning they were either aborted due to timeouts or exceeded available resources before compilation completed. The center panel plots actual compilation times for successful runs: depth-1 configurations are fast (7–31s), while depth-4 and depth-5 reach 100s and 357s respectively, reflecting the combinatorial explosion of TCAM entries as tree depth increases. The right panel shows the estimated pipeline stage count for each compiled configuration, with the hard cap of 16 stages marked in red. Depth-1 forests fit comfortably between 8 and 20 stages as tree count increases; depth-2 ranges from 11 to 17; depth-3 reaches 14–18; and depth-4 exceeds 17 stages even for the smallest configurations. This confirms that the 20-stage pipeline limit is the dominant hard constraint for in-network Random Forest deployment, and that configurations with depth > 3 combined with more than 4 trees are not viable on the current Tofino 2 hardware.

The compilation status grid highlights the operational boundaries of in-network ML. Configurations that exceed a certain complexity threshold either time out during compilation or fail completely due to the exhaustion of active pipeline stages and logical table IDs.

5.2.3. Feature Importance and Resource Overview

To optimize the models for deployment, it is crucial to understand which features contribute most to the classification. The Gini feature importances confirm that a small subset of the MARINA telemetry features provides the majority of the predictive power.

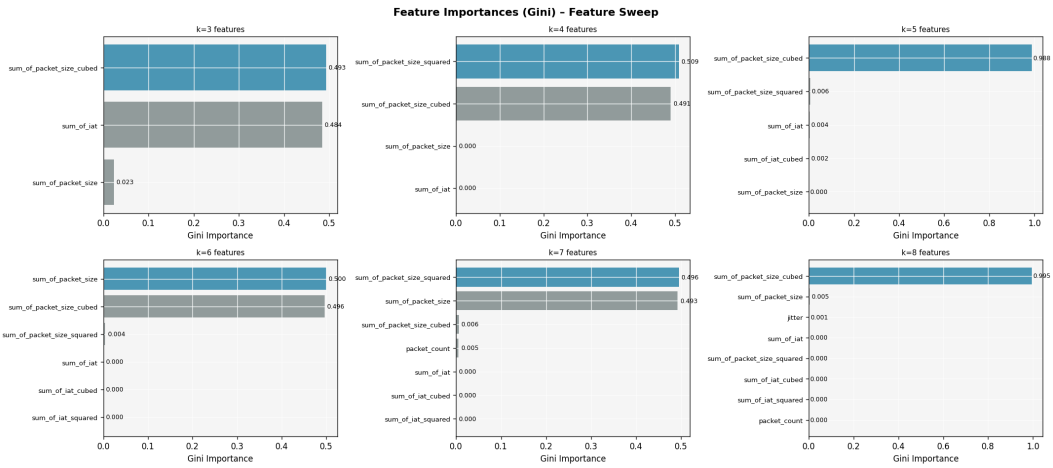

Figure 16: **Feature importances calculated during the feature sweep experiment.**

Figure 16 presents the Gini importance scores for each feature subset tested during the feature sweep. At $k = 3$, the model splits its decision-making power almost equally between sum_of_packet_size_cubed (0.493) and sum_of_iat (0.484), with sum_of_packet_size contributing only 0.023. At $k = 4$, sum_of_packet_size_squared becomes the dominant feature (0.509), with sum_of_packet_size_cubed close behind (0.491), while both sum_of_packet_size and sum_of_iat drop to essentially zero importance. A striking pattern emerges at $k = 5$: sum_of_packet_size_cubed alone captures 0.988 of the total Gini importance, effectively collapsing the model into a near-single-feature classifier. This extreme concentration suggests that the third-order moment of the packet size distribution encodes the most discriminative signal for buffer depletion prediction. At $k = 6$ and $k = 7$, the importance redistributes more evenly between sum_of_packet_size and its higher-order moments (0.50 each), with all other features near zero. At $k = 8$, sum_of_packet_size_cubed again dominates at 0.995, with jitter (0.001) and sum_of_packet_size (0.005) playing only marginal roles. Across all configurations, IAT-based features (sum_of_iat, sum_of_iat_squared, sum_of_iat_cubed) and packet_count consistently score at or near zero, confirming that throughput volume statistics are far more predictive of QoE state than timing-based features in this dataset.

The aggregated resource overview illustrates the complex multi-dimensional trade-off between test accuracy, total SRAM allocated, active pipeline stages, and VLIW instructions. It proves that by carefully tuning the hyperparameters, an optimal configuration can be found that maximizes QoE prediction accuracy while remaining strictly within the hardware bounds of the Tofino 2 ASIC.

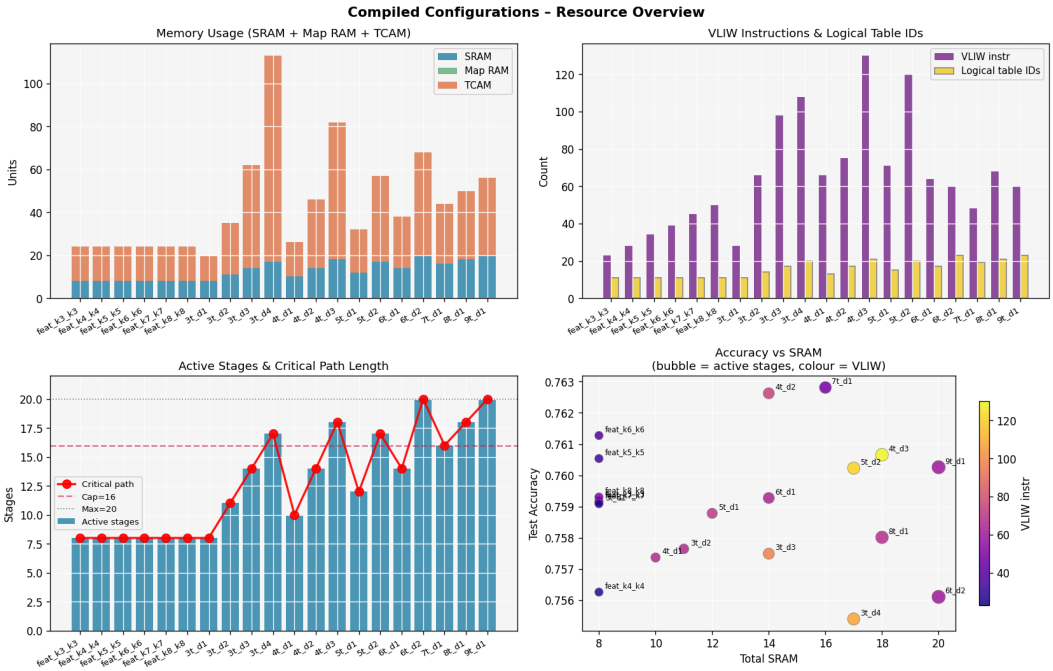

Figure 17: **Combined resource overview mapping accuracy against SRAM usage and active stages.**

Figure 17 consolidates all successfully compiled configurations into a unified resource view. The top-left memory panel shows that feature-sweep configurations (feat_k3 through feat_k8) consume a uniformly low SRAM and TCAM footprint of roughly 24 units total, while tree/depth sweep configurations scale substantially: 4t_d4 peaks at approximately 112 units of combined SRAM and TCAM. TCAM dominates the memory budget in deeper configurations because each additional tree depth doubles the number of ternary match entries required per stage. The top-right panel shows that VLIW instruction counts and logical table IDs track the number of trees and depth closely: configurations such as 4t_d3 and 5t_d2 exceed 100 VLIW instructions, creating significant pressure on the instruction word budget. The bottom-left panel plots active pipeline stages and the critical-path length across all configurations. Feature-sweep entries cluster at 8 active stages regardless of k, while deeper tree configurations progressively grow toward the 20-stage maximum. The critical path, marked in red, frequently exceeds the 16-stage soft cap for configurations with depth $g = 3$, explaining the compiler timeouts observed in the previous analysis. Finally, the bottom-right scatter plot maps test accuracy against total SRAM, with bubble size encoding active pipeline stages and colour encoding VLIW instruction count. The Pareto-optimal region – highest accuracy at lowest resource cost – is occupied by configurations feat_k6 (accuracy 0.761, SRAM 8) and 4t_d2 (accuracy 0.763, SRAM 14). Configurations such as 3t_d4 (accuracy 0.756, SRAM 16) and 6t_d2 (accuracy 0.756, SRAM 21) represent poor trade-offs, consuming significantly more resources for lower or equal accuracy. This analysis confirms that shallow forests (depth 1–2) with a moderate number of trees (4–7) and a small, high-importance feature set (k=4–6) represent the practical sweet spot for deployable in-network QoE classification on the Tofino 2 ASIC.

5.3. Conclusion

This work bridges the gap between high-level machine learning models and low-level data plane programming, demonstrating that Quality of Experience prediction from encrypted video traffic is not only theoretically possible, but practically achievable within the strict physical constraints of modern programmable ASICs.

5.3.1. Summary of Contributions

The central hypothesis of this thesis, that a constrained Random Forest classifier mapped to Ternary Match-Action Tables can perform accurate, line-rate QoE prediction within Intel Tofino, was confirmed by the experimental results. Three concrete contributions support this conclusion:

First, the MARINA feature pipeline was successfully adapted for in-network deployment. By reducing the feature space to four hardware-feasible metrics (`ps_sum`, `ps2_sum`, `ps3_sum`, and `jitter`) and aggregating them over non-overlapping 50ms windows, the system achieved a classification accuracy of 0.76 (15 percentage points above the majority-class baseline) with an F1-score of 0.80 for the critical **At_Risk** class. This demonstrates that a small, carefully selected feature subset can retain the majority of predictive power while remaining deployable on hardware.

Second, the pForest architecture was successfully extended and adapted to the Tofino Native Architecture (TNA). The automated pipeline translates a trained scikit-learn Random Forest directly into compilable P4 code, unrolling each decision tree across different physical pipeline stages and encoding split thresholds as ternary bitmasks in TCAM. The voting logic is pre-computed and hardcoded as exact-match entries, eliminating the need for costly runtime arithmetic in the data plane. This end-to-end automation significantly reduces the effort required to deploy new ML models onto programmable switches.

Lastly, the sweep across tree count, depth, and feature count revealed that the 20-stage pipeline limit of the Tofino 2 architecture is the dominant bottleneck, not SRAM or TCAM. Configurations with depth $d > 3$ and more than six trees consistently exceeded the stage budget and failed to compile.

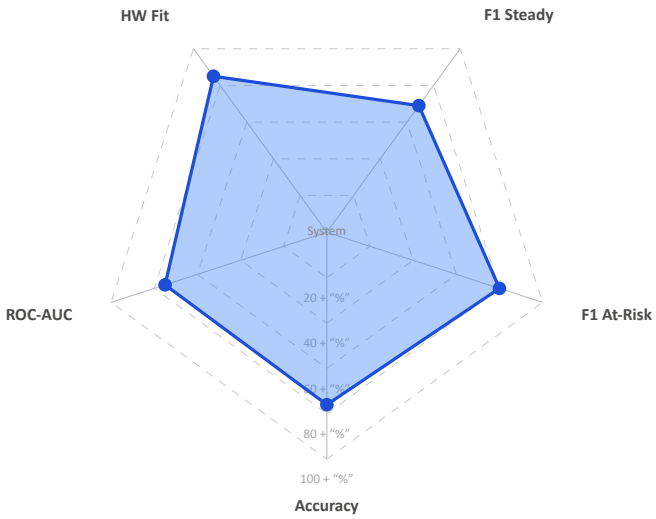

Figure 18: **System performance overview across key evaluation dimensions. Hardware Fit reflects the ratio of successful compilations within the 20-stage limit; Pipeline Efficiency reflects stage utilization headroom; ROC-AUC reflects the model's ability to distinguish between the two classes.**

5.3.2. Interpretation of Results

The feature importance analysis consistently confirmed that higher-order statistical moments of traffic volume, `ps2_sum` and `ps3_sum`, contribute more than 44% of the model's decision-making power. This finding aligns with the broader literature on encrypted traffic analysis: because payload content is inaccessible under HTTPS and QUIC, the **shape** of the traffic distribution, captured by its second and third moments, encodes information about burstiness and asymmetry that correlates strongly with application-layer buffer behavior. Jitter, while contributing only 11.7% of Gini importance in the baseline model, remains a necessary feature for capturing transient network instability that precedes stalling events.

The risk-based labeling strategy, which uses a 500ms lookahead window to define the **At_Risk** class, proved to be a meaningful design choice. The resulting class imbalance (61.3% **At_Risk** vs. 38.7% **Steady**) reflects the nature of the dataset: video streaming sessions frequently operate near buffer depletion thresholds, particularly under the automatic quality selection of scenario 6.

The hardware-in-the-loop evaluation confirmed that the Tofino switch correctly computed stateful flow metrics, including packet counts, IAT sums, and jitter, entirely within the data plane, without offloading any computation to an external server.

5.3.3. Limitations and Future Work

Despite these results, several limitations should be acknowledged. The model was trained and evaluated on a single scenario (scenario 6, automatic quality selection) of the Würzburg QoE dataset. The dataset contains 7 other scenarios (Scenarios 1-5, 7, 8) that introduce various network degradations like rate-limiting and delay. Evaluating on these rate-limited scenarios would likely induce distribution shifts; specifically, rate-limited scenarios trigger more extreme buffer drops, meaning more **At_Risk** events and potentially

inflated recall. Generalizability to other platforms (e.g., Netflix, Twitch), transport protocols (e.g., QUIC-based streaming), or network environments (e.g., high-latency satellite links) remains to be validated.

Furthermore, the deployment of a single unified Random Forest introduces a trade-off between model simplicity and classification precision during the early phases of a flow, where feature statistics have not yet converged. Future work could explore context-dependent models that use a lightweight early-phase classifier until sufficient flow statistics have accumulated, transitioning to the full model thereafter.

Finally, the current implementation does not yet close the loop between the Translator's classification output and active network management decisions. Integrating the QoE prediction result into a dynamic traffic shaping or prioritization policy, for example, by signaling the control plane to adjust queue weights for flows classified as **At_Risk**, would complete the vision of a fully autonomous, in-network QoE management system.

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List of Abbreviations

AI **– Artificial Intelligence** 1

ASIC **– Application-Specific Integrated Circuit** 2, 7, 8, 18, 28, 31, 32

BFRT **– Barefoot Runtime** 3, 19, 19, 19, 19, 21, 38

BWE **– Bandwidth Estimate** 16

DAG **– Directed Acyclic Graph** 4, 8

DTA **– Data-Plane Telemetry Architecture** 6, 6, 6, 7, 7, 7, 7, 22, 22, 22, 22, 22, 23, 24, 24, 24, 25, 26, 26, 38, 38, 39

EMA **– Exponential Moving Average** 10, 10, 10, 10, 10, 11, 23

IAT **– Inter-Arrival Time** 3, 14, 15, 15, 23, 25, 26, 34

IP **– Internet Protocol** 1, 19, 21, 21, 23, 26, 38

ITU **– International Telecommunication Union** 1

KPI **– Key Performance Indicator** 1, 2

MAC **– Media Access Control** 4, 19, 20, 20, 21, 21, 21, 21, 23, 23, 23, 23, 23, 23, 24, 26, 38

MAT **– Match-Action Table** 3, 7, 17

ML **– Machine Learning** 2, 1, 1, 1, 3, 3, 3, 4, 4, 4, 19, 20, 22, 23, 27, 27, 27, 28, 29, 30, 33, 38, 38

PCAP **– Packet Capture** 20, 22, 22, 23, 23, 24, 39

PSNR **– Peak Signal-to-Noise Ratio** 2, 3

QoE **– Quality of Experience** 2, 2, 2, 2, 2, 2, 2, 2, 3, 1, 1, 1, 1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 3, 6, 9, 10, 11, 11, 11, 12, 20, 22, 22, 22, 24, 26, 26, 31, 31, 32, 33, 34, 35, 35, 38, 39, 39

RDMA **– Remote Direct Memory Access** 7, 7, 38, 38

RF **– Random Forest** 6, 27

SDN **– Software-Defined Networking** 18

SSIM **– Structural Similarity Index** 2, 3

***TCAM* – Ternary Content-Addressable Memory** 3, 4, 4, 4, 5, 8, 8, 18, 28, 28, 28, 30, 32, 32, 32, 33, 33, 38

***UDP* – User Datagram Protocol** 7, 23, 23, 23

***UHD* – Ultra-High-Definition** 1, 38

***VQA* – Video Quality Assessment** 1, 2